John C. Flournoy, PhD

Research Scientist: Measurement, Evaluation & Statistical Inference

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Education

Ph.D., Psychology, University of Oregon, 2018
M.S., Psychology, University of Oregon, 2014
B.A., Cognitive Science, University of California at Berkeley, 2005

Academic Appointments

- Associate of the Department of Psychology (Courtesy Appointment), Harvard University, 2024 - Present

Research

Independent Research and Data Science Consultant

2018 - Present

- Clients include: Somerville Lab at Harvard (HCP-D neuroimaging), True Life Center (mental health), and Meru Health (digital mental health)

Developer Success Lab, Pluralsight — *Principal Research Scientist*

2024 - 2025

- Co-designed a structured interview protocol and ran 20 interviews with infrastructure engineers (SREs, cloud developers) on how information and voice move through organizations. Began qualitative coding of interviews.
- Designed randomized survey-embedded experiments testing whether brief interventions changed developers' self-efficacy for learning. One framed testing as a tool for learning and memory rather than evaluation; the other prompted developers to use communities of practice.
- Led analysis of ticket data from 11k+ developers using Bayesian multilevel models: commonly cited levers showed small associations with cycle time, and variation within and between developers means a single measurement is low signal (No Silver Bullets, Empirical Software Engineering, 2025).
- Founded the lab's community-engagement overlay journal (dsl.pubpub.org).

Harvard University, Cambridge, MA — *Research Associate*

2020 - 2024 (Postdoctoral Fellow, 2018 - 2020)

- Lead analyst on a dense-sampling fMRI study; quantified the test-retest reliability of task-fMRI measures (lead author, NeuroImage 2024).
- Co-designed intensive longitudinal modeling of stress, sleep, and digital communication leading to 3 publications. NIMH grant R37-MH119194 ($9,034,169)
- Analyzed multi-site, longitudinal fMRI data of adolescent cognitive control and reward (NIMH grant U01-MH109589; $17,141,357).
- Provided methodology and scientific computing support to 8 graduate students, 15 post-bac RAs, and 9 post-docs, across 2 lab groups

Skills

- Bayesian modeling
- Causal inference
- Collaborative science
- Cloud computing
- Data visualization
- Experimental design
- Git and GitHub
- Human participants research
- Machine learning
- Measurement reliability & validity
- Missing data
- Multilevel modeling
- Psychometrics
- Qualitative coding
- R, Stan, Python, SQL
- Structural Equation Modeling
- Structured interview & protocol design
- Reproducible science

Tools & Packages

`claude-analysis-config`: Claude Code reviewer agents and a tested hook that catch errors in analysis code.

`riclpmr`: generate syntax for random intercept cross-lag panel models.

`curvish`: Bayesian analysis and visualization of GAM smooths using 1st and 2nd derivatives.

`scorequaltrics`: retrieve and score data from Qualtrics using csv templates.

Special Training

ICPSR Summer Program in Quantitative Methods, 2015

- Causal Inference for the Social Sciences
- Advanced Bayesian Models for the Social Sciences

University of Oregon, Eugene, OR — *Graduate Research Fellow*
2012 - 2018

- Developed Bayesian model of reinforcement learning to measure adolescent social motives and associations with health risking behavior.
- Led data collection for 300+ participants including foster-care-involved youth
- Analyzed longitudinal personality and fMRI task data using MLM & SEM.
- Validated new self report and task measures

Stanford University, Stanford, CA — *Research Coordinator*
2009 - 2012

- Coordinated Simons Foundation study examining sleep in autism
- Site coordinator, registered clinical trial of novel PET marker in dementia

Selected Publications

See google scholar for the full list of more than 25 peer-reviewed articles

Flournoy, J. C., Lee, C. S., Wu, M., & Hicks, C. M. (2025). No Silver Bullets: Why Understanding Software Cycle Time is Messy, Not Magic. *Empirical Software Engineering*, *30*(6), 174

Flournoy, J. C., Bryce, N. V., Dennison, M. J., Rodman, A. M., McNeilly, E. A., Lurie, L. A., ... & McLaughlin, K. A. (2024). A precision neuroscience approach to estimating reliability of neural responses during emotion processing: Implications for task-fMRI. *NeuroImage*, *285*, 120503.

Bryce, N., **Flournoy, J.C.**, Moreira, J. F. G., Rosen, M. L., Sambook, K. A., Mair, P., & McLaughlin, K. A. (2021). Brain parcellation selection: An overlooked decision point with meaningful effects on individual differences in resting-state functional connectivity. *NeuroImage*, 118487.

Flournoy, J. C., Vijayakumar, N., Cheng, T. W., Cosme, D., Flannery, J. E., & Pfeifer, J. H. (2020). Improving practices and inferences in developmental cognitive neuroscience. *Developmental cognitive neuroscience*, 100807.

Ludwig, R. M., **Flournoy, J. C.**, & Berkman, E. T. (2019). Inequality in personality and temporal discounting across socioeconomic status? Assessing the evidence. Journal of research in personality, 81, 79-87.

Matta, T. H., **Flournoy, J. C.**, & Byrne, M. L. (2018). Making an unknown unknown a known unknown: Missing data in longitudinal neuroimaging studies. *Developmental cognitive neuroscience*, 33, 83-98.

Selected Teaching & Talks

July 2025. Guest, *How to do your job happier*. The Stack Overflow Podcast.

November, 2021. *Why and How to Care About Covariates in Longitudinal Data*. Presented at the 7th Annual UC Adolescence Consortium Institute.

May 2021. *Scientific Practice in Developmental Cognitive Neuroscience*. Presented at the Lifespan Informatics and Neuroimaging Center, Department of Psychiatry, University of Pennsylvania Perelman School of Medicine.